

Preserving What Matters: Semantic Scaffolds Beyond Saturation in Summarization Evaluation

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Abstract

Summarization ships in countless production systems, making model selection a routine decision that depends on measuring summary quality. Existing metrics struggle to support this: ROUGE captures only surface overlap, while LLM-as-judge scores saturate to near-identical values that fail to rank models effectively. We observe this saturation across three public datasets, two proprietary datasets, and multilingual settings. Motivated by this, we introduce **Semantic Scaffold**, an evaluation framework that extracts a hierarchical representation of facts, questions, and entity attributes from a source text, labeling each as a main point or supporting detail, and reusing this structure as a fixed reference for scoring summaries. From this representation, we derive three diagnostic metrics: Fact Preservation Score (**FPS**), Question Preservation Score (**QPS**), and Entity Preservation Score (**EPS**), designed to reward the preservation of essential information while penalizing detail overload, and position them as interpretable diagnostics that remain informative where holistic axes collapse. Finally, we analyze four recurring failure modes of ROUGE and LLM-as-judge scores, demonstrating that scaffold-based evaluation remains informative where conventional metrics collapse.

1 Introduction

Document summarization, a long-standing NLP challenge (Lin, 2004; See et al., 2017), is increasingly deployed in production settings, powering everything from search overviews to automated digests of reports. This ubiquity makes model selection a routine production decision that hinges on accurately measuring summary quality.

Existing evaluation methods fall short. ROUGE (Lin, 2004) and BLEU (Papineni et al., 2002) count n-gram overlap, missing semantic content in abstractive summaries. Human evaluation is reliable but costly at scale (Fabbri et al., 2021). LLM-as-judge scoring (Liu et al., 2023a; Zheng et al., 2023) is cheaper but poorly calibrated (Fabbri et al., 2021), hard to interpret, and biased toward verbose text (Zheng et al., 2023); moreover, automatic metrics in general struggle to reliably evaluate LLM-generated summaries (Goyal et al., 2022; Zhang et al., 2024). On our internal Contracts dataset, the Hallucination axis spans only 0.00225 across eight models, while the widest holistic-axis span is 0.182 (Table 4). We call this loss of range near the top of the scale *metric saturation*; compressed scores do not imply identical model quality.

We propose the **Semantic Scaffold**, a proactive evaluation framework that addresses these limitations (Figure 1). The scaffold is a structured representation of a document’s key facts, questions, hierarchy, and entities, extracted once from the source for reuse. Our

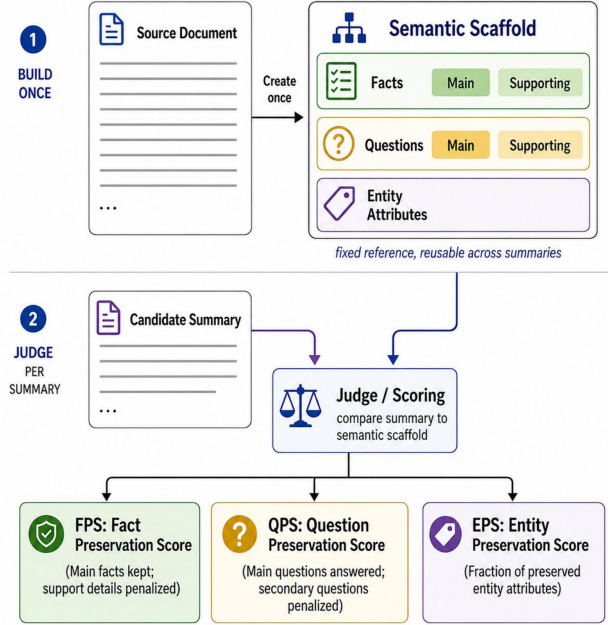


Figure 1: Overview of the Semantic Scaffold for summarization evaluation.

decoupled “extract-then-judge” approach uses this hierarchy to distinguish between main facts/questions and supporting details. This enables a modular toolbox of interpretable metrics that specifically reward summaries for preserving the document’s core structure by prioritizing main points over less critical details.

Contributions.

- (1) We document substantial LLM-judge score compression across proprietary, public, and multilingual summarization settings;
- (2) we introduce the **Semantic Scaffold**, a reusable source representation that enables item-level evaluation of fact, question, and entity preservation; and
- (3) we catalogue four recurring failure modes of the ROUGE plus LLM-judge stack, each paired with the model-selection decision it corrupts.

2 Related Work

Reference-based. Lexical-overlap metrics, ROUGE (Lin, 2004), BLEU (Papineni et al., 2002), and METEOR (Banerjee & Lavie, 2005), rely on surface lexical matching and miss meaning and factual accuracy (Fabbri et al., 2021). Embedding metrics such as BERTScore (Zhang et al., 2020) and MoverScore (Zhao et al., 2019) capture deeper similarity but still need references.

Reference-free. Fact-based metrics and their meta-evaluations (Gabriel et al., 2021), including SummaC (Laban et al., 2022), FactCC (Kryściński et al., 2020), and FineSurE (Song et al., 2024), flag factual errors against the source. QA-based metrics such as QAEval (Deutsch et al., 2021) and QuestEval (Scialom et al., 2021) generate questions and check whether the summary answers them, but couple question generation to scoring, so the questions cannot be reused. LLM-as-judge methods (Zheng et al., 2023) like G-Eval (Liu et al., 2023a) rate fluency, coherence, and factuality; they track human judgments but remain poorly calibrated (Fabbri et al., 2021) and length-biased (Zheng et al., 2023).

Dataset ¹	DS range	Hall. range
ArXiv-Sum	9.01 to 9.42	9.99 to 10.00
GovReport	9.00 to 9.34	9.97 to 10.00
BigPatent	9.23 to 9.55	10.00 to 10.00

Table 1: Direct-scoring (DS) saturation on three public benchmarks, five models each. The largest unrounded DS spread is 0.412; every Hallucination mean is at least 9.972.

From atomic units to structure. The Pyramid Method (Nenkova & Passonneau, 2004) scored summaries through atomic content units; Liu et al. (2023b) formalized this as the ACU protocol and AutoACU (Liu et al., 2023c) automated it. Later work adds factual verification and alignment: FActScore (Min et al., 2023), ACUEval (Wan et al., 2024), MiniCheck (Tang et al., 2024), AlignScore (Zha et al., 2023), and HAMLET (Lee et al., 2025); notably, FActScore finds frontier LLMs below 60% atomic factual precision in long-form generation. The Semantic Scaffold extends this line with explicit main/supporting hierarchy, a representation spanning facts, questions, and entities, and a single extraction reused across models.

3 Saturation: When LLM Judges Stop Ranking Models

Our goal was not to study metric saturation. However, when evaluating newer and cheaper models for production use cases, we discovered saturation in our LLM-as-a-judge scores, diminishing the usefulness of these metrics for selecting the top model. This section establishes this observation across proprietary production data, three public benchmarks and multilingual settings.

Dataset. We use two annotated internal datasets and three public datasets. **Screenshots:** 235 bug-report UI screenshots selected from items with three available human references, with one reference retained per instance (Figure 5); short, structured, and entity-dense. **Contracts:** 445 documents with dense legal language and multi-thousand-word inputs, where omissions and structural compression matter. We also evaluate on three public long-document benchmarks: **ArXiv-Sum** (Cohan et al., 2018) (6,440 papers), **GovReport** (Huang et al., 2021) (973 reports), and **BigPatent** (Sharma et al., 2019) (8,565 patents across nine CPC shards).

Models. We evaluate eight frontier models on the proprietary datasets: GPT-4o-mini, GPT-4.1-mini, GPT-4o, GPT-4.1, Gemini-2.0-Flash, Gemini-2.5-Flash, Gemini-2.5-Pro, and Claude-3.7-Sonnet. For the **Screenshots** dataset, we additionally include a *Human* reference row as a target anchor.

We denote the three proprietary systems as Internal-S, Internal-M, and Internal-L. For public benchmark evaluation, we use five models: Internal-S, Internal-L, GPT-4.1-mini, Gemini-2.5-Flash, and Claude-3.7-Sonnet; the qualitative walkthrough in Appendix B also includes Internal-M.

Observed saturation. We evaluate every summary with a G-Eval-style (Liu et al., 2023a) LLM judge on four holistic axes (Faithfulness, Completeness, Understandability, and Hallucination; prompt in §G.7).

Proprietary datasets. On Contracts, every Hallucination mean rounds to 1.00; the unrounded span is 0.00225. Faithfulness spans 0.057 and Understandability 0.025. Screenshots shows similar compression: Hallucination spans 0.025 and Understandability 0.040. Per-model scores are in Table 4.

Public benchmarks. Saturation is not confined to our internal data. On three widely used public benchmarks scored on a 1–10 scale, the widest DS spread across five models is 0.412, and Hallucination is at least 9.972 in every cell (Table 1).

¹ArXiv-Sum (Cohan et al., 2018), GovReport (Huang et al., 2021), BigPatent (Sharma et al., 2019).

Model	Lang. group	Faith. avg	Hall. avg
Gemini-2.5-Flash	English	9.67	9.99
	Non-English	9.63	9.99
GPT-4.1-mini	English	9.57	9.97
	Non-English	9.55	9.97
Claude-3.7-Sonnet	English	9.73	9.99
	Non-English	9.66	9.98

Table 2: Cross-lingual saturation on three internal screenshot datasets (Faith./Hall. averaged over English vs. nine other languages).

Multilingual evidence. Across three models, three datasets, and ten languages (90 aggregate cells), Hallucination ranges from 9.95 to 10.00 (Table 2). OCR text and judge prompts were professionally translated into nine languages with human review. These data show compression in this translated pipeline, not a causal effect of language. Per-language breakdowns are in Appendix C.

4 Semantic Scaffold

Saturation is a measurement failure, the score stops discriminating exactly when generation quality and decision stakes are highest. Motivated by this, we introduce the Semantic Scaffold.

4.1 Architecture

The scaffold has three properties: It is **proactive**, with evaluation targets fixed before seeing any summary to prevent criteria drift; **decoupled**, as extraction runs once while judgment runs per candidate; and **reusable**, with one scaffold serving all summaries of a source for consistent, cheap comparisons.

Two stages. *Creation* uses one LLM pass to extract a structured representation of facts, questions, hierarchy, and entities from the source. *Judgment* then scores each candidate against this fixed reference. Each score links to a scaffold item, providing item-level evidence of what a summary kept, dropped, or misrepresented, rather than a single holistic verdict.

Components. Each scaffold item is tagged a *main point* or a *supporting detail*. **Atomic facts** are concise, self-contained statements (20–100 per document). **Key questions** capture what the document answers, each with a brief expected answer and source-sentence provenance. **Entity attributes** list salient entities (people, organizations, products, concepts) and their defining properties for entity-level checks. Templates are in Appendix G.

4.2 Baseline Metrics

Our method builds on FineSurE (Song et al., 2024), adopting its two-stage structure and its **Fact Faithfulness (F)** and **Fact Coverage (FC)** metrics. We add **Question Coverage (QC)** as a third baseline (Table 3). However, FineSurE lacks hierarchy, treating all facts equally. Our scaffold metrics extend it by tagging all facts and questions as main or supporting and rewarding summaries that prioritize the main tier. Appendix A summarizes all metric definitions; formulas for FPS, QPS, and EPS follow below.

²Built on FineSurE (Song et al., 2024).

Metric	Formula	What it measures
F	faithful / total facts	Fact faithfulness ²
FC	covered / total facts	Fact coverage ²
QC	answered / total questions	Question coverage

Table 3: Baseline evaluation metrics.

4.3 Scaffold Metrics

Fact Preservation Score (FPS). FPS rewards keeping main facts and penalizes supporting ones.

$$\text{FPS} = \frac{M_s}{M_c} \times \left(1 - \frac{S_s}{S_c}\right) \quad (1)$$

where M_s and S_s denote main and supporting facts preserved in the summary, and M_c and S_c denote main and supporting facts in the scaffold. The first term measures the share of main facts preserved; the second term, $(1 - S_s/S_c)$, penalizes inclusion of supporting facts. Reproducing every detail is closer to paraphrase than summary; stating all supporting facts ($S_s = S_c$) drives FPS to 0 regardless of main-fact coverage.

Question Preservation Score (QPS). QPS is the question analogue, rewarding answers to main questions and penalizing attention to secondary ones.

$$\text{QPS} = \frac{Q_s}{Q_c} \times \left(1 - \frac{R_s}{R_c}\right) \quad (2)$$

with Q_s and R_s denoting main and supporting questions answered in the summary, and Q_c and R_c denoting main and supporting questions in the scaffold.

Entity Preservation Score (EPS). Entity fidelity is especially important in domains such as legal, medical, and financial text, where a dropped attribute changes meaning. Unlike FineSurE’s binary entity check, EPS is continuous, the fraction of scaffold entity attributes preserved:

$$\text{EPS} = \frac{|\text{preserved attributes}|}{|\text{scaffold attributes}|} \quad (3)$$

so it distinguishes summaries that keep some attributes from those that keep none.

Key Point: Summary goals drive metric choice

The three scaffold metrics are not interchangeable; each metric supports a different summary goal. *EPS leads when entity context must be preserved* (e.g. contracts, financial/medical text, identifier-dense screenshots). *QPS leads when the summary must answer main questions* (e.g. triage notes, executive digests). *FPS leads when a compressed abstract summary is required* (e.g. briefings, news-style digests).

Implementation. We use GPT-4.1 for both extraction and judgment (temperature 0.1, max_tokens = 4096) and score models zero-shot in their default configurations. A single model builds the scaffold and judges against it so that every candidate is scored against an identical reference; this is what makes the scaffold reusable across models and comparisons cheap.

5 Main Results

Table 4 reports all metrics across two proprietary datasets, Contracts and Screenshots.

Dataset	Model	Faith.	Compl.	Underst.	Hall.	F [†]	FC [†]	FPS [‡]	QC	QPS [‡]	EPS
Screenshots	GPT-4o-mini	0.89	0.82	0.96	0.97	0.93	0.43	0.51	0.36	0.73	0.79
	GPT-4.1-mini	0.93	0.87	0.97	0.99	0.97	0.49	0.52	0.42	0.70	0.87
	Gemini-2.0-Flash	0.96	0.86	0.98	1.00	0.98	0.48	0.51	0.42	0.69	0.85
	Gemini-2.5-Flash	0.95	0.87	0.97	1.00	0.99	0.53	0.53	0.45	0.67	0.88
	GPT-4o	0.87	0.77	0.95	0.98	0.97	0.31	0.39	0.25	0.70	0.71
	GPT-4.1	0.92	0.85	0.97	0.99	0.97	0.47	0.52	0.39	0.71	0.86
	Gemini-2.5-Pro	0.95	0.89	0.98	0.99	0.97	0.55	0.50	0.47	0.65	0.85
	Claude-3.7-Sonnet	0.97	0.90	0.99	1.00	0.98	0.57	0.47	0.50	0.62	0.90
<i>Human</i>		0.95	0.86	0.98	0.99	0.91	0.53	0.40	0.44	0.60	0.76
Contracts	GPT-4o-mini	0.92	0.81	0.96	1.00	0.98	0.14	0.39	0.19	0.60	0.28
	GPT-4.1-mini	0.97	0.89	0.96	1.00	0.98	0.20	0.50	0.28	0.75	0.46
	Gemini-2.0-Flash	0.95	0.82	0.96	1.00	0.98	0.15	0.41	0.21	0.61	0.30
	Gemini-2.5-Flash	0.95	0.84	0.97	1.00	0.95	0.15	0.42	0.21	0.63	0.31
	GPT-4o	0.95	0.71	0.96	1.00	1.00	0.06	0.16	0.09	0.27	0.14
	GPT-4.1	0.98	0.86	0.97	1.00	0.99	0.15	0.43	0.22	0.64	0.38
	Gemini-2.5-Pro	0.93	0.83	0.98	1.00	0.95	0.15	0.42	0.21	0.63	0.31
Claude-3.7-Sonnet		0.97	0.87	0.98	1.00	0.96	0.17	0.46	0.24	0.67	0.36

Table 4: Descriptive scores on both proprietary datasets. On Contracts, Hallucination spans 0.002, Understandability 0.025, Faithfulness 0.057, and Completeness 0.182; EPS spans 0.32. The strongest value per column is highlighted in green; pink marks saturated cells. The **Human** row (Screenshots only) is an anchor, not a ceiling. [†]Song et al. (2024)

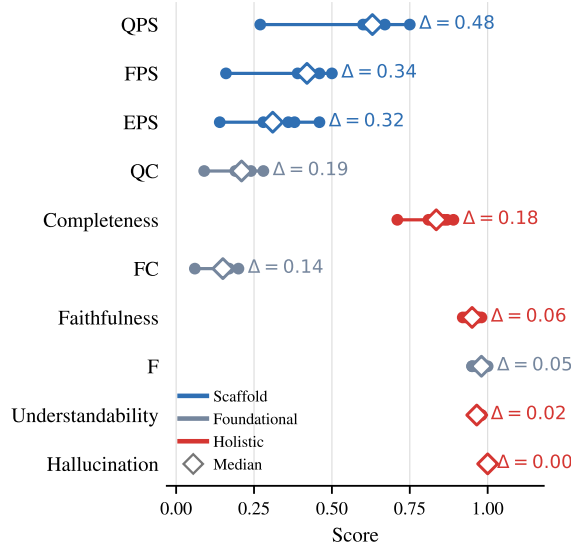


Figure 2: Score range across 8 models on Contracts, sorted by spread (dots = models). Holistic metrics (red) saturate near 1.0; scaffold metrics (blue) retain spread; foundational (gray) in between.

Scaffold metrics restore discrimination. Where holistic metrics saturate (§3), scaffold metrics maintain separation: QPS, FPS, and EPS span $\Delta = 0.32$ to 0.48 on Contracts, compared to Hallucination ($\Delta = 0.00$), Understandability (0.02), and Faithfulness (0.06) (Figure 2). This represents a wider spread, making model differences visible where holistic axes collapse. Even when baseline metrics like FC and QC cannot differentiate (e.g., 0.14 and 0.19 spreads),

scaffold metrics retain resolution, keeping model differences visible where holistic axes no longer distinguish systems. We report spread as resolving power; pairing it with a preference target is the natural next step for using these axes to select models.

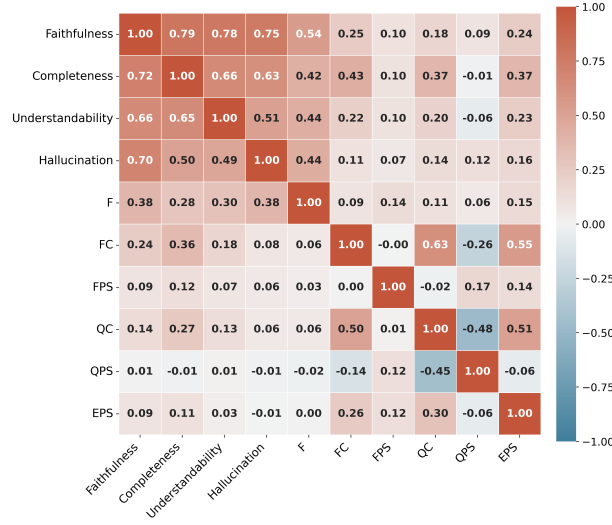


Figure 3: Pairwise correlations of evaluation metrics on the Screenshots dataset. The upper triangle reports Spearman’s ρ ; the lower triangle reports Pearson’s r .

Themes are easy; hierarchy is hard. A systematic gap between QPS and FPS scores reveals a core limitation in current LLMs. Across every model and dataset, QPS exceeds FPS (e.g., GPT-4.1-mini on Contracts: QPS = 0.75 vs. FPS = 0.50). The two criteria are calibrated differently by design: question alignment credits a question whenever the summary supplies enough information to answer it, while fact alignment requires the fact itself to be conveyed (SG.5). The gap therefore locates where a summary’s coverage sits between thematic and propositional—models track a document’s agenda more readily than its fact hierarchy—and raising FPS at fixed QPS and EPS is a well-posed target under matched criteria.

FPS, QPS, and EPS are only weakly correlated with one another (Figure 3), confirming they measure distinct aspects of summary quality.

The negative Pearson QPS/QC correlation ($r = -0.45$) reflects their differing objectives: QC rewards breadth of question coverage, while the $(1 - R_s/R_c)$ term in QPS explicitly discounts it in favour of main questions.

A small model leads on two scaffold axes. On Contracts, GPT-4.1-mini outscores every larger model on both QPS (0.75) and EPS (0.46) while sitting at parity on the holistic axes. A team trusting holistic scores alone would pay for a bigger model that is structurally worse, over-provisioning compute for a task the smaller, cheaper model already wins. Saturation is thus not only an evaluation problem; it translates directly into wasted inference cost and weaker production decisions.

Humans aren’t the ceiling; they’re the target. Seven of eight models exceed the Human row on EPS (Human = 0.76). EPS is recall-oriented over scaffold attributes, an objective annotators are not writing toward, so the human row functions as a *sensible-target* anchor rather than a ceiling, and is best read alongside the holistic axes rather than as a standalone quality verdict.

Scaffold metrics on public benchmarks, under grouped summarization, are reported in Table 11 and Figure 6.

6 Failure-Mode Analysis

Saturation is the most visible failure of holistic evaluation, but not the only one. By constructing grouped multi-document settings from single-document benchmarks (Mukherjee et al.,

ID	Name	Effect	Production consequence
F1	Catastrophic compression collapse	Hallucination drops from near-perfect (10.0) to 3.6 - 7.0 on the same source when the judge conflates omission with fabrication	Good summaries look hallucinated
F2	Ranking instability	ROUGE-L and Direct Scoring select different best models in all four tested (dataset, setting) configurations	Metric choice, not model quality, decides what ships
F3	Domain-driven ROUGE bias	ROUGE-L scores are approximately 2× higher on government reports than earnings calls for summaries of comparable quality	Cross-domain comparison becomes invalid
F4	Goodhart rule-breaking	A model violating the 100-word budget achieves the highest grouped direct score; prompt relaxation fails to correct the behavior	Non-compliant summaries get rewarded

Table 5: Four failure modes in grouped multi-document evaluation, each paired with the model-selection decision it corrupts.

Dataset	Set.	ROUGE-L leader	DS leader
ECT+FindSum	individual	Claude-3.7-Sonnet (.111)	GPT-4.1-mini (9.40)
GovRep+BillSum	individual	Gemini-2.5-Flash (.228)	GPT-4.1-mini (9.44)
ECT+FindSum	grouped	GPT-4.1-mini (.085)	Internal-L (6.81)
GovRep+BillSum	grouped	GPT-4.1-mini (.119)	Internal-L (6.69)

Table 6: F2 (**ranking instability**): ROUGE-L and Direct Scoring select different metric leaders in all four (dataset, setting) configurations. No independent preference target establishes which leader has higher quality. ECT = ECTSum (Mukherjee et al., 2022), FindSum (Liu et al., 2022), GovRep = GovReport (Huang et al., 2021), BillSum (Kornilova & Eidelman, 2019).

2022; Kornilova & Eidelman, 2019; Huang et al., 2021), we observe four failure patterns in our data, F1 to F4 (Table 5). These are descriptive stress tests, not isolated causal experiments. We summarize them here; details are in Appendix D.

F1 (Catastrophic collapse under compression). When five models compress related documents into a 100-word budget, Direct Scoring falls from 9.1–9.4 individually to 4.2–6.8 grouped, and Hallucination decreases by about 5 points on average (Figure 4). The cause is not fabrication but a measurement artifact: the judge compares the 100-word summary against a much longer reference, so any detail dropped to meet the budget looks “unsupported.”

The next three failures concern the rankings these metrics produce. **F2 (instability):** ROUGE-L and Direct Scoring crown different top models in all four (dataset, setting) configurations we tried (Table 6); with disagreement that complete, it is the choice of metric, not the quality of the model, that decides which system ships. **F3 (domain bias):** for summaries of comparable quality, ROUGE-L runs about 2× higher on government reports than on earnings calls, so a score is only meaningful within a domain and cross-domain comparisons are not valid. **F4 (Goodhart):** the model that most often breaks the 100-word limit earns the highest grouped Direct Score, and relaxing the limit only lowers ROUGE-L without fixing the behavior.

Why hierarchical scaffolds help. Hierarchical scaffold addresses this by scoring components independently. On the grouped dataset, scaffold metrics sustain separation (Supporting Fact Coverage: 3.4×, Main Fact Coverage: 1.8×; Appendix D.3, Table 11), exposing the real culprit: the Hallucination collapse isn’t fabrication, models drop supporting details

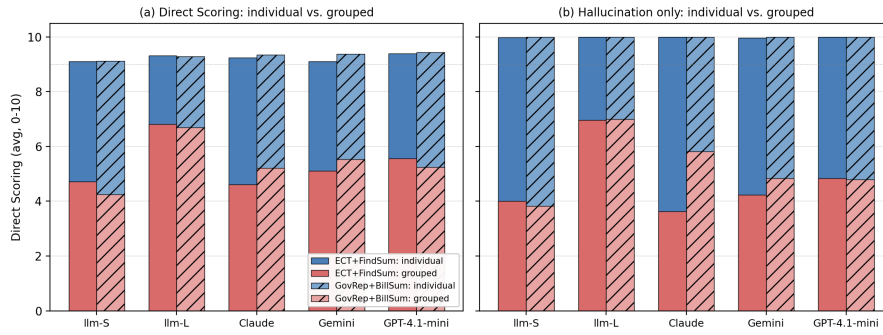


Figure 4: **F1: compression collapse** on two multi-document benchmarks, individual (*blue*) versus grouped 100-word (*red*). Direct Scoring and Hallucination both decrease

under compression. While holistic scores flag that something’s wrong, scaffolds reveal the cause.

7 Conclusion

When every frontier model earns a near-perfect hallucination score, the bottleneck has shifted from the models to the way we measure them. We argued that holistic LLM-judge metrics lose discriminative power exactly as generation quality rises, and we showed it across two proprietary datasets, three public benchmarks, and ten languages, then cataloged four recurring failure modes of the conventional stack.

Our response, the **Semantic Scaffold**, changes the structure of evaluation rather than the prompt: it extracts a hierarchical reference from the source once and scores every candidate against it, so the scores trace back to specific facts, questions, and entities. Where holistic axes flatten, its FPS, QPS, and EPS metrics still resolve differences between models, and under heavy compression they pinpoint *why* a model fails instead of collapsing into uninterpretable numbers.

Takeaway: when every model scores near the ceiling, the task is probably not solved; the metric has stopped resolving, and that is the moment to switch to structural, decomposed evaluation.

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Limitations and Future Work

The main limitation of our framework is its reliance on the quality of the initial semantic scaffold. All downstream metrics depend on the accuracy and completeness of the LLM-based extraction. Therefore, a key direction for future work is to develop stronger methods to improve and automatically assess scaffold quality, especially in specialized or noisy domains. Further research will also explore broader uses of the Semantic Scaffold, such as content recommendation, educational assessment, and cross-lingual analysis.

Two extensions would strengthen the framework’s use as a selection criterion. First, a preference-based meta-evaluation: Screenshots retains three human references per item, making it well suited to relating scaffold scores to human preference and to calibrating the main/supporting labels against human salience judgments. Second, broader reporting: pairwise comparison is the standard non-saturating alternative to direct scoring, and situating the scaffold metrics against it, with length normalization and interval estimates, would clarify where each is preferable.

Ethics Statement

The proprietary **Screenshots** and **Contracts** datasets were annotated in-house under standard employment agreements; no crowd-sourcing was used. Source documents and scaffolds exclude end-user personally identifiable information (PII), and entity annotations cover organizations, products, system identifiers, and document-internal references rather than personal attributes.

Because scaffold construction and scoring are LLM-assisted, the resulting metrics should be interpreted as comparative evaluation signals rather than absolute guarantees of summary quality, especially in sensitive use cases.

Metric	Measures	Description
Baseline		
LLM-as-Judge	Holistic Quality	LLM rates summary on faithfulness, completeness, coherence, relevance. Susceptible to biases.
Foundational: Fact-Based		
Fact Faithfulness (F)	Fact Accuracy	Proportion of key facts accurately represented.
Fact Coverage (FC)	Completeness	Proportion of key facts present in summary.
Main Fact Coverage (Main FC)	Main-point Recall	M_s / M_c , the proportion of main facts present in the summary.
Supporting Fact Coverage (Supp. FC)	Supporting-detail Recall	S_s / S_c , the proportion of supporting facts present in the summary.
Fact Focus (F. Focus)	Factual Precision	Proportion of summary sentences aligned to at least one scaffold fact.
Foundational: Question-Based		
Question Coverage (QC)	Completeness	Proportion of key questions answered in summary.
Showcase Metrics: Hierarchy		
Fact Preservation (FPS)	Information Hierarchy	Preserves main facts while avoiding overemphasis on supporting details.
Question Preservation (QPS)	Hierarchical QA	Answers main questions without excessive focus on supporting ones.
Entity Preservation (EPS)	Attribute Accuracy	Correctly preserves entity-attribute pairs from source.

Table 7: Evaluation metrics in the Semantic Scaffold framework.

A Semantic Scaffold Evaluation Framework: Metric Overview

Table 7 summarizes the key metrics used in our framework, organized into Baseline, Foundational, and Showcase categories. The Foundational metrics (Section 4, Table 3) assess basic factual accuracy and coverage, while the Showcase metrics (FPS, QPS, EPS) evaluate the higher-order structural and entity-level qualities that remain informative once holistic LLM-judge scores saturate. The subsequent sections give the exact prompt templates for each.

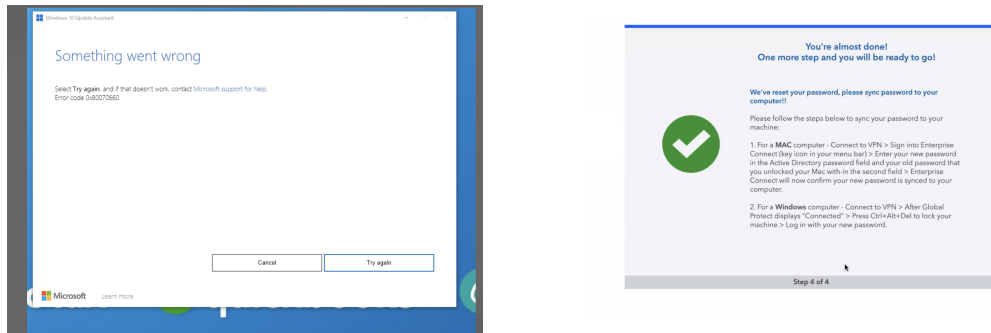


Figure 5: **Screenshot** examples from the proprietary dataset: a Windows Update error dialog (*left*) and a password-sync instruction page (*right*). Error codes, UI labels, and step-by-step identifiers drive summary quality, making **EPS** differences salient.

B Qualitative Walkthrough: From Source to Scaffold to Scores

This appendix walks one example (internal-dataset-v1/task-0099) from raw OCR text through scaffold construction to final scores, as a concrete reference for reproducing the pipeline.

B.1 Source Document (OCR)

Camera. Allow access to the camera on this device. If you allow access, people using this device will be able to choose if their apps have camera access by using the settings on this page. Denying access blocks Microsoft Store apps and most desktop apps from accessing the camera. It does not block Windows Hello. Camera access for this device is on. Change. Allow apps to access your camera. If you allow access, you can choose which apps can access your camera by using the settings on this page. Denying access blocks apps from accessing your camera. It does not block Windows Hello. On. Some desktop apps may still be able to access your camera when settings on this page are off. Find out why.

These 108 OCR words from a Windows 10 privacy-settings screenshot pack a two-level permission hierarchy, two exceptions (*Windows Hello* and desktop apps), a state (access enabled), and an action (“Find out why”). It is a typical **Screenshots** sample: short and entity-rich, where small differences in factual and entity preservation separate strong summaries from adequate ones and where holistic metrics blur exactly what the scaffold keeps.

B.2 Extracted Scaffold

The scaffold organizes the document into three views (atomic facts, key questions, and entity attributes), each item labeled *main point* or *supporting detail*.

Atomic Facts (main points first, followed by supporting details).

- M1. Camera access on the device can be enabled or disabled. (*main point*)
- M2. When device-level access is enabled, users can choose which individual applications are permitted camera access. (*main point*)
- M3. Denying device-level access blocks Microsoft Store applications and most desktop applications from accessing the camera. (*main point*)
- M4. Camera access for this device is currently enabled. (*main point*)
- S1. Denying camera access does not block *Windows Hello*. (*supporting detail*)
- S2. Some desktop applications may still access the camera even when per-application settings are disabled. (*supporting detail*)
- S3. A “Find out why” link is provided to explain the desktop application exception. (*supporting detail*)

Key Questions (main questions first, followed by supporting details).

- Q1. What functionality does enabling device-level camera access provide to users? (*main point*)
- Q2. What happens to application-level camera access when device-level access is denied? (*main point*)
- Q3. What is the current camera access state for this device? (*main point*)
- R1. Does denying camera access affect *Windows Hello*? (*supporting detail*)
- R2. Can desktop applications still access the camera when per-application settings are disabled? (*supporting detail*)

Entity Attributes

- **Camera access (device level)**
 - Can be enabled or disabled
 - Currently enabled
 - Enables per-application access control when active

Model	Faith.	Hall.	FPS	EPS
Internal-S	0.92	0.98	0.33	0.44
Internal-M	0.90	0.97	0.22	0.33
Internal-L	0.93	0.98	0.38	0.56
GPT-4o	0.94	0.99	0.25	0.67

Table 8: Illustrative per-instance scores for task-0099 (approximate; final values depend on sentence-level alignments produced by the judge). Holistic metrics (Faithfulness, Hallucination) stay clustered across all four summaries, while FPS and EPS separate them.

- **Camera access (application level)**
 - Individual applications may be permitted or blocked
 - Per-application settings are available only when device-level access is enabled
- **Windows Hello**
 - Not blocked by camera access denial
 - Operates independently of the privacy settings page
- **Desktop applications**
 - Blocked when device-level access is denied
 - May still access the camera despite per-application settings being disabled

B.3 Model Summaries and Per-Instance Scores

We compare four models on the same instance. All four are broadly faithful and coherent, yet they differ in how well they preserve the document’s structure, entity attributes, and exceptions.

Internal-S (189 characters):

“The document explains camera access settings for a device, allowing users to choose which apps can access the camera, with a note that denying access blocks most apps but not Windows Hello.”

Internal-M (173 characters):

“Camera access settings for this device and apps are detailed, with options to allow or deny access. Some desktop apps may still access the camera even when settings are off.”

Internal-L (162 characters):

“The document explains camera access settings, allowing or denying access to the camera for apps, with notes on exceptions for Windows Hello and some desktop apps.”

GPT-4o (223 characters):

“The document explains camera access settings on a device, allowing users to control app access to the camera. It notes that denying access blocks most apps except Windows Hello and mentions exceptions for some desktop apps.”

B.4 Step-by-Step FPS and EPS Computation

We walk through Internal-S to show that scaffold scores trace to specific semantic units rather than an opaque judgment.

FPS. The scaffold has $M_c = 4$ main facts (M1–M4) and $S_c = 3$ supporting facts (S1–S3). The summary covers M2 (“choose which apps can access the camera”) and M3 (“denying access blocks most apps”), partially signals M1, and omits M4, so $M_s = 2$; it mentions S1 (“not Windows Hello”) but not S2 or S3, so $S_s = 1$. By Equation 1, $FPS = \frac{2}{4} \left(1 - \frac{1}{3}\right) \approx 0.33$.

Model	Dataset	Metric	en	de	es	fr	it	ja	ko	nl	pt	pt-BR
Gemini-2.5-Flash	Dart-v1	F	9.69	9.63	9.62	9.64	9.66	9.67	9.63	9.67	9.61	9.61
		H	9.98	9.99	9.98	9.99	10.00	9.99	9.99	10.00	9.99	9.99
	Dart-qe-v1	F	9.67	9.61	9.58	9.60	9.61	9.65	9.57	9.64	9.59	9.62
		H	9.99	9.98	9.99	9.98	10.00	9.99	9.98	9.98	9.99	9.99
	Screen-v1	F	9.65	9.64	9.62	9.63	9.60	9.64	9.58	9.65	9.62	9.62
		H	9.99	9.99	9.99	9.99	9.99	9.98	9.99	9.99	9.99	9.99
GPT-4.1-mini	Dart-v1	F	9.55	9.47	9.54	9.57	9.57	9.56	9.53	9.57	9.56	9.54
		H	9.95	9.95	9.97	9.98	9.97	9.96	9.97	9.98	9.98	9.97
	Dart-qe-v1	F	9.55	9.53	9.54	9.56	9.57	9.56	9.52	9.57	9.54	9.55
		H	9.97	9.98	9.97	9.98	9.96	9.96	9.97	9.97	9.97	9.96
	Screen-v1	F	9.59	9.55	9.54	9.57	9.58	9.56	9.53	9.58	9.57	9.54
		H	9.98	9.97	9.97	9.98	9.97	9.97	9.97	9.97	9.97	9.97
Claude-3.7-Sonnet	Dart-v1	F	9.73	9.67	9.65	9.66	9.67	9.65	9.64	9.67	9.66	9.65
		H	9.98	9.97	9.99	9.98	9.99	9.98	9.99	9.98	9.98	9.98
	Dart-qe-v1	F	9.76	9.70	9.68	9.67	9.68	9.67	9.65	9.68	9.67	9.67
		H	9.99	9.97	9.98	9.98	9.99	9.99	9.99	9.98	9.98	9.99
	Screen-v1	F	9.70	9.63	9.65	9.65	9.66	9.64	9.63	9.66	9.65	9.64
		H	9.99	9.97	9.98	9.98	9.98	9.98	9.99	9.98	9.98	9.98

Table 9: Full cross-lingual evaluation results across three screenshot datasets and ten languages. F = Faithfulness; H = Hallucination.

GPT-4o covers more main facts ($M_s = 3$) but reproduces more supporting detail ($S_s = 2$), giving $FPS = \frac{3}{4}(1 - \frac{2}{3}) = 0.25$: despite higher raw coverage it scores lower, because the metric rewards hierarchy, not volume.

EPS. The scaffold holds 9 entity attributes across 4 entities. The small model preserves 2/3 device-level attributes, 1/2 application-level, 1/2 for Windows Hello, and 0/2 for desktop applications, so $EPS = 4/9 \approx 0.44$, matching the automated judge.

Key takeaway. All four summaries get nearly identical holistic scores (Faithfulness ≥ 0.90 , Hallucination ≥ 0.97), yet FPS spans about 0.16 and EPS about 0.34 on this single instance. Such per-instance gaps compound across the dataset into the larger separations in Table 4.

C Full Cross-Lingual Saturation Results

§3 reports cross-lingual results averaged across datasets. This appendix gives the full per-language breakdown for all three evaluated models and datasets.

Evaluation setup. We test multilingual robustness on three internal screenshot datasets: **Screen-v1** (887 images), **Dart-v1** (831), and **Dart-qe-v1** (215). OCR text was professionally translated into nine languages with human review. Summaries used the same production prompt as English; to avoid English-pivot bias, the judge prompt was translated too, so each summary is scored against its target-language source. This measures end-to-end multilingual quality, not translation fidelity alone.

Key observations. Saturation persists across languages: within each model family, Faithfulness varies by at most 0.12 and Hallucination by at most 0.07 across ten languages, far less than the between-model gaps on English. All Hallucination scores fall within $[9.95, 10.00]$, a far narrower band than the 3.6 to 7.0 collapse under grouped summarization, so severe compression, not multilinguality, is the dominant production failure mode.

D Failure-Mode Analysis: Full Results

D.1 F1: Compression Collapse Across Models and Datasets

Table 10 reports the complete set of (model, dataset, setting) results underlying Figure 4.

Dataset	Model	Setting	Faith.	Compl.	Underst.	Hall.	Avg
ECT+FindSum	Internal-S	ind.	8.95	7.85	9.66	9.98	9.11
		grp.	3.93	3.96	6.99	4.00	4.72
	Internal-L	ind.	9.44	8.08	9.77	10.00	9.32
		grp.	5.90	6.25	8.15	6.97	6.82
	Claude-3.7-Sonnet	ind.	9.08	8.06	9.83	10.00	9.24
		grp.	3.75	4.05	6.99	3.63	4.60
	Gemini-2.5-Flash	ind.	8.96	8.01	9.50	9.97	9.11
		grp.	4.26	4.71	7.26	4.24	5.12
	GPT-4.1-mini	ind.	9.44	8.40	9.76	10.00	9.40
		grp.	4.55	5.29	7.54	4.84	5.55
GovReport+BillSum	Internal-S	ind.	9.07	7.82	9.62	10.00	9.13
		grp.	3.48	3.19	6.48	3.83	4.24
	Internal-L	ind.	9.42	8.06	9.68	10.00	9.29
		grp.	5.86	6.01	7.90	7.00	6.69
	Claude-3.7-Sonnet	ind.	9.22	8.33	9.85	10.00	9.35
		grp.	3.93	4.00	7.09	5.81	5.21
	Gemini-2.5-Flash	ind.	9.36	8.38	9.74	10.00	9.37
		grp.	4.84	4.97	7.50	4.83	5.53
	GPT-4.1-mini	ind.	9.56	8.50	9.70	10.00	9.44
		grp.	4.30	4.66	7.23	4.79	5.24

Table 10: Complete individual-to-grouped evaluation results for F1 (**compression collapse**). Best-performing model per (dataset, setting) is shown in **bold**.

Reading guide. *Individual* (ind.) generates one unconstrained summary per document; *grouped* (grp.) generates a single 100-word summary per task group. The source documents are identical across settings, isolating the effect of aggressive compression.

Key observations. The full table confirms the pattern from Figure 4 at the per-cell level: every model collapses under the grouped constraint, and Hallucination falls hardest (for example, Claude-3.7-Sonnet from 10.00 to 3.63 on ECT+FindSum). Internal-L records the highest grouped scores despite weaker individual summaries, the Goodhart effect detailed under F4.

Why Hallucination collapses: judge-compression conflation. The observed collapse is not primarily caused by fabricated claims. Manual inspection of 30 low-scoring grouped outputs reveals a more subtle failure mode.

In the grouped setting, the judge compares a 100-word summary against a reference formed by concatenating two single-document summaries (approximately 450 to 600 words total). Consequently, factual details omitted for compression are often treated as unsupported.

For example, an abstractive statement such as “the CNMI permit program was amended” may be penalized when the reference instead contains a detailed legal formulation such as “Public Law No. 110-229, enacted in 2008, amended the U.S.-CNMI covenant.” Although semantically consistent, the compressed version lacks the statutory detail expected by the judge.

The result is a systematic conflation of *compression loss* with *fabrication*. By operating over atomic semantic units rather than surface text overlap, the **Semantic Scaffold** separates these effects: missing secondary details manifest as a *Supporting Fact Coverage gap*, rather than an artificial collapse in Hallucination.

Rank	Model	F. Faith.	F. Cov.	F. Focus	Main FC	Supp. FC	FPS	QPS	EPS
1	Claude-3.7-Sonnet	0.986	0.539	0.752	0.693	0.316	0.453	0.615	0.801
2	Gemini-2.5-Flash	0.961	0.493	0.866	0.674	0.260	0.481	0.655	0.789
3	GPT-4.1-mini	0.975	0.444	0.835	0.627	0.216	0.476	0.693	0.773
4	Internal-S	0.807	0.287	0.992	0.400	0.103	0.353	0.628	0.596
5	Internal-L	0.726	0.288	1.000	0.385	0.092	0.343	0.649	0.598
Best-to-worst spread		0.260	0.252	0.248	0.308	0.224	0.138	0.078	0.205
Relative spread		1.4×	1.9×	1.3×	1.8×	3.4×	1.4×	1.1×	1.3×

Table 11: Scaffold-based metrics averaged across grouped multi-document benchmark settings.

D.2 F2: Ranking Instability, Full Per-Setting Breakdown

Table 6 in the main paper summarizes the four (dataset, setting) configurations in which ROUGE-L and Direct Scoring (DS) select different top-performing models. Here, we provide the complete per-setting breakdown underlying that disagreement.

Why the rankings diverge. In the individual setting Claude-3.7-Sonnet tops ROUGE-L: its more extractive style overlaps more with the references. GPT-4.1-mini tops Direct Scoring with more abstractive summaries the judge rates higher on faithfulness and coherence. In the grouped setting Internal-L leads Direct Scoring but not ROUGE-L, the same F4 non-compliance effect. Because the disagreement holds in every configuration, metric choice alone, not noise, decides which system ships.

D.3 Scaffold Metrics Under Grouped Summarization

Figure 6 shows that scaffold-based metrics retain meaningful separation even when holistic Direct Scoring collapses under grouped summarization. Table 11 provides the complete metric breakdown averaged across both grouped benchmark datasets.

Column interpretation. Faithfulness (**F. Faith.**) stays high, so the grouped drop comes from omission, not fabrication, while overall fact coverage (**F. Cov.**) falls sharply under the 100-word cap. **Main Fact Coverage** spreads more widely (1.8×) than total coverage, so stronger models keep the *right* facts rather than simply more of them, and **Supporting Fact Coverage** spreads widest (3.4×): supporting detail is the first casualty of compression. **FPS** combines main-point recall with a supporting-detail penalty, and **QPS** and **EPS** stay comparatively stable while still separating models.

Compared with the proprietary results, the ordering of spreads inverts here: Main and Supporting Fact Coverage separate models most (1.8× and 3.4×), while FPS, QPS, and EPS compress (0.138, 0.078, 0.205 best-to-worst). This is expected under a uniform 100-word cap, which pushes every system toward the same operating point on the main/supporting trade-off that FPS and QPS are built to expose. On this data the hierarchy-aware components are the informative axes and the aggregates are best read as summaries of them.

A revealing failure signature. Internal-L reaches perfect F. Focus (1.000), meaning every sentence matches some scaffold fact, yet records the lowest Supporting Fact Coverage (0.092): it stays on-topic by filtering to a narrow slice of content instead of synthesizing across documents.

D.4 F4: Prompt Relaxation Does Not Recover Performance

A natural response to the compression collapse observed in F1 is to relax the summarization constraint itself. If the 100-word cap is driving the degradation, then allowing longer and more flexible summaries should restore evaluation quality.

To test this hypothesis, we compare two grouped summarization prompts:

Prompt v1 (constrained summarization). *“Generate a concise summary (less than 100 words) that captures the main topic, purpose, and key details of the documents...”*

Prompt v2 (production-oriented summarization). *“Generate a summary of these documents as it would appear attached to a record in a service management platform. Focus on the information most useful for a service agent or knowledge worker to understand the document’s purpose, key findings, and any actionable outcomes without reading the full document. Use clear, plain language proportional to the document’s complexity.”*

Table 12 compares the two settings. Prompt v1 uses the original grouped 100-word cap, while v2 removes the length constraint and instead relies on a production-style summarization objective.

Dataset	RG-L v1 (v1)	RG-L v2 (v2)	DS v1 (v1)	DS v2 (v2)
Dart-v1	0.358	0.275	8.99	8.86
Dart-qe-v1	0.355	0.251	8.84	8.77
GovReport+ BillSum	0.228	0.102	6.69	5.80
ECT+FindSum	0.111	0.090	6.82	5.76

Table 12: Prompt comparison: v1 (100-word cap) versus v2 (no explicit cap; production-oriented instruction), averaged across five models.

Key observations. The grouped Direct Score leader is lower under v2 on all four datasets (for example, 6.69 to 5.80 on GovReport+BillSum and 6.82 to 5.76 on ECT+FindSum), and the individual ROUGE-L leader also decreases. Because v2 changes both framing and the length instruction, these results do not isolate either factor or show that prompt wording cannot recover performance.

E Metric Design Rationale and Edge Cases

E.1 Why a Multiplicative Form for FPS and QPS

FPS and QPS multiply a main-point recall term (M_s/M_c) by a supporting-detail penalty ($1 - S_s/S_c$) rather than adding them. The product imposes a *joint* constraint: a summary scores well only if it both preserves central content *and* compresses secondary detail. An additive form would not do this: a weighted sum can still reward a summary that copies every supporting fact, or under-penalize one that drops core content. With the product, failure on either axis pulls the score toward zero: no main-fact coverage gives FPS=0 regardless of supporting detail, and reproducing all supporting facts gives FPS=0 regardless of recall. This encodes the basic assumption of summarization: keep the priority structure, compress the rest.

FPS is most informative for abstractive, length-constrained tasks where the goal is to preserve a document’s informational hierarchy: bug-report digests, executive briefings, legal case summaries, news. It is a poor standalone target for regulated or high-recall workflows (clinical trial protocols, regulatory filings, audit reports) where retaining supporting detail is desirable; there, read FPS alongside FC and Supporting Fact Coverage.

E.2 EPS: Continuous Rather Than Binary Entity Evaluation

FineSurE flags entity consistency with a binary, sentence-level criterion: a sentence either has an entity error or it does not. That misses the gradation in practice, where a summary may name an entity correctly yet drop, compress, or distort some of its attributes. EPS instead works at the attribute level, measuring the fraction of scaffold-defined entity attributes preserved:

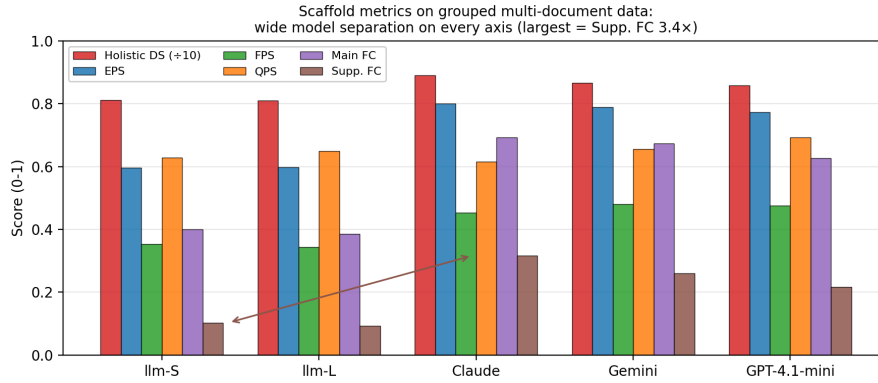


Figure 6: Scaffold metrics on grouped multi-document summarization (five models). Each axis keeps meaningful spread (Supporting Fact Coverage $3.4\times$, Main Fact Coverage $1.8\times$, EPS $1.3\times$, FPS $1.4\times$) while holistic Direct Scoring (*red*, shown as DS/10) compresses into 0.81 to 0.89. The widest spread, Supporting Fact Coverage, matches the F1 diagnosis: under compression, supporting details fail first.

- A summary that mentions “Windows Hello” but omits its independence from the privacy-settings toggle preserves only 1/2 attributes for that entity.
- A summary that mentions an entity name but misattributes all associated properties contributes 0, equivalent to omitting the entity entirely.
- A summary preserving all entity attributes across the scaffold receives EPS = 1.0.

This granularity matters in legal, technical, and enterprise settings, where an entity carries many important attributes (contract clauses, error codes, system components, legal references) and partial preservation differs meaningfully from omission.

E.3 Reporting hierarchy without an inverse penalty

Main and supporting coverage can be reported separately without assuming that supporting content should reduce quality. Designing a validated hierarchy-aware aggregate requires held-out evaluation and remains future work.

F Failure-Mode Figures: Extended Captions and Guided Reading

Figure 6 plots the scaffold metrics on the same grouped data as Figure 4 in the main text.

In Figure 4, Direct Scoring and Hallucination both decrease in the grouped setting. Internal-L degrades least and also under-complies with the 100-word limit, but these data do not establish that non-compliance causes the difference. Figure 6 localizes coverage differences without identifying their cause.

G Complete Prompt Templates with Design Notes

All scaffold extraction and evaluation prompts are provided verbatim. Each prompt is prefixed with a design note explaining the key choices.

Common settings for all prompts: model = GPT-4.1, temperature = 0.1, max_tokens = 4096, response_format: json_object. The low temperature (0.1 rather than 0.0) prevents perfectly deterministic but occasionally degenerate JSON while keeping scores highly reproducible.

G.1 Fact Extraction

Design note. The 20–100 range spans short documents (screenshots yield 8–15 facts) and long ones (contracts yield 60–100). The three-tier label routes metadata and boilerplate to “other” so it cannot affect scores, and ordering main points first doubles as a consistency check on the extraction.

Fact Extraction Prompt

System: You are a precise information extraction assistant.
Return valid JSON only. No preamble, no explanation.

User:

Extract 20–100 key facts from the document below.

Each fact must be:

- Concise (at most 2–3 entities per fact)
- Self-contained (understandable without the full document)
- Labeled with exactly one of three types:
 - "main point" -- central claim, finding, or action
 - "supporting detail" -- context, evidence, elaboration
 - "other" -- metadata, dates, boilerplate

Order: all main points first, then supporting details, then other.
Do not include duplicate or redundant facts.

Output JSON format:

```
{
  "key_facts": [
    { "id": "1", "fact": "...", "type": "main point" },
    { "id": "2", "fact": "...", "type": "supporting detail" },
    ...
  ]
}
```

Document:
{{DOCUMENT_TEXT}}

G.2 Question Extraction

Design note. Questions capture the document’s implied agenda. The expected answer and source-sentence provenance make the scaffold auditable and let the question-alignment step (§G.5) align at the sentence level without re-reading the full document.

Question Extraction Prompt

System: You are a precise information extraction assistant.
Return valid JSON only. No preamble, no explanation.

User:

Extract 20–100 key questions from the document below.

For each question provide:

- A brief expected answer (1–2 sentences, from the document)
- A type: "main point", "supporting detail", or "other"
- The source (e.g. "Sentence 3" or "Paragraph 2")

Order: main-point questions first, then supporting, then other.

Output JSON format:

```
{
  "questions": [
    {
      "id": "1",
      "question": "...",
      "answer": "...",
      "type": "main point",
      "source": "Sentence 3"
    },
    ...
  ]
}
```

Document:
{{DOCUMENT_TEXT}}

G.3 Entity Attributes Extraction

Design note. Entity extraction is deliberately broad (people, organizations, products, legal instruments, technical and abstract concepts), and each attribute is a single statement so the entity-preservation prompt (§G.6) can check it independently.

Entity Extraction Prompt

System: You are a precise information extraction assistant.
Return valid JSON only. No preamble, no explanation.

User:
Extract all key entities from the document below.

Entities include: people, organisations, products, concepts, legal instruments, technical terms, or any named object central to the document.

For each entity list its key attributes. Each attribute must:

- Be a single concise factual statement (one claim)
- Be labeled: "main point", "supporting detail", or "other"

Output JSON format:

```
{
  "entities": [
    {
      "entity_name": "...",
      "entity_type": "person | org | product | concept | other",
      "attributes": [
        {"attribute": "...", "type": "main point"},
        {"attribute": "...", "type": "supporting detail"}
      ]
    }
  ]
}
```

Document:
{{DOCUMENT_TEXT}}

G.4 Key Fact Alignment Evaluation

Design note. This prompt computes M_s and S_s in the FPS formula (Equation 1 in the main paper) by determining which scaffold facts are inferable from the summary. The “sentence IDs” output enables item-level audit: a practitioner can inspect exactly which summary sentences support each scaffold fact, identifying precisely which main-point facts were missed. The justification field discourages shallow keyword matching.

Fact Alignment Prompt

System: You are a precise evaluation assistant.
Return valid JSON only. No preamble, no explanation.

User:
You receive a document summary and key facts from the original document.

For each key fact, determine if it can be inferred from the summary. “Inferred” means the fact’s meaning is clearly conveyed, even in different words. Do NOT mark a fact as inferred if the summary is only tangentially related.

For inferred facts: list the summary sentence ID(s) that support it, and explain why in 1 sentence.

Include all facts in the output, even those not inferred.

Output JSON format:

```
{
  "fact_alignment": [
    {
      "fact_id": "1",
      "inferred": true,
      "sentence_ids": [2],
      "justification": "Sentence 2 states that..."
    },
    {
      "fact_id": "3",
```

```

    "inferred": false,
    "sentence_ids": [],
    "justification": "No sentence in the summary covers this."
  }
]
}

```

Key facts (with types):
 {{FACTS_JSON}}

Summary (sentences numbered starting at 1):
 {{NUMBERED_SUMMARY}}

G.5 Key Question Alignment Evaluation

Design note. Structurally analogous to §G.4 but operating on questions. A question is “answered” if the summary provides enough information for a reader to answer it, not if the question is literally asked and answered. This softer criterion is why QPS tends to exceed FPS across models.

Question Alignment Prompt

System: You are a precise evaluation assistant.
 Return valid JSON only. No preamble, no explanation.

User:
 You receive a document summary and key questions about the original document.

For each question, determine if the summary provides enough information for a reader to answer it. If yes, list the sentence ID(s) containing the relevant information and explain in 1 sentence.

Output JSON format:

```

{
  "question_alignment": [
    {
      "question_id": "1",
      "answered": true,
      "sentence_ids": [1, 3],
      "justification": "..."
    },
    {
      "question_id": "4",
      "answered": false,
      "sentence_ids": [],
      "justification": "The summary does not address..."
    }
  ]
}

```

Key questions (with types):
 {{QUESTIONS_JSON}}

Summary (sentences numbered starting at 1):
 {{NUMBERED_SUMMARY}}

G.6 Entity Preservation Evaluation

Design note. Checks attribute-by-attribute preservation rather than entity-level presence, enabling the continuous EPS score described in Appendix E. Only attributes that appear in the summary are returned; the EPS denominator is always the total scaffold attribute count regardless of what the summary mentions.

Entity Preservation Prompt

System: You are a precise evaluation assistant.
 Return valid JSON only. No preamble, no explanation.

User:
 You receive a document summary and a list of key entities with their attributes, extracted from the original document.

For each entity that appears in the summary, list the attributes

that are correctly and faithfully represented. An attribute is "correctly represented" if its meaning is preserved in the summary, even in different words.

Do NOT include entities absent from the summary.
Do NOT include attributes that are distorted or only partially correct.
Do NOT invent attributes not listed in the input.

Output JSON format:

```
{
  "entity_preservation": [
    {
      "entity_name": "...",
      "preserved_attributes": [
        "Exact text of the preserved attribute from the input list"
      ]
    }
  ]
}
```

Entities and attributes from the scaffold:
{{ENTITIES_JSON}}

Summary to evaluate:
{{SUMMARY_TEXT}}

G.7 Direct Score Evaluation (holistic baseline)

Design note. This is the holistic LLM-as-judge prompt used for all direct-scoring baselines. The 1 to 10 scale with explicit anchors at 1, 5, and 10 matches prior work (SummEval, FineSurE) and provides meaningful gradations. Hallucination is framed as its *absence* (10 = no hallucination) so higher is always better across all four dimensions. The judge receives only the raw source document, not the scaffold, making it a true holistic baseline without hierarchy information. The rationale field aids human auditing of low-scoring cells.

Direct Score (Holistic Baseline) Prompt

System: You are a rigorous and impartial evaluation assistant.
Return valid JSON only. No preamble, no explanation.

User:
Evaluate the quality of the summary against the original document. Rate on four dimensions using a 1-10 scale where 10 is perfect.

Dimension definitions:

1. Faithfulness
 - 10 = perfectly faithful; no distortions or omissions of information that is present
 - 5 = mostly faithful; minor inaccuracies
 - 1 = fabricates or contradicts the document
2. Completeness
 - 10 = covers all key information needed for understanding
 - 5 = covers some key points but misses important ones
 - 1 = misses most key information
3. Understandability
 - 10 = perfectly clear, logical, easy to follow
 - 5 = somewhat clear; has clarity or organisation issues
 - 1 = incomprehensible or severely disorganised
4. Hallucination (absence of hallucination)
 - 10 = no hallucination; all claims supported by the source
 - 5 = some claims not clearly supported; questionable
 - 1 = severe fabrications; significant unsupported claims

Output JSON format:

```
{
  "faithfulness": N,
  "completeness": N,
  "understandability": N,
  "hallucination": N,
  "rationale": "1-2 sentence explanation of the scores"
}
```

Original document:
{{DOCUMENT_TEXT}}

Summary to evaluate:
{{SUMMARY_TEXT}}